CHANDU PRASAD BHAIRAPU



AI Engineer

bhairapuchanduprasad@gmail.com (+49) 15560194877 Cologne • EU Blue Card linkedin github

ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in taking ML and AI solutions into production: containerized services with Docker and Kubernetes, CI/CD pipelines, model validation and monitoring, on AWS and Microsoft Azure. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by a strong foundation in machine learning, data analysis and Python development across enterprise and start-up environments.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Sole developer owning the full knowledge-extraction AI pipeline end-to-end – data ingestion, chunking, prompt engineering, and indexing (agentic, deterministic, and monolithic strategies) – as Python services on Docker, Kubernetes and Azure, reporting directly to the CTO and working with SAP domain experts.
- Designed and optimized multi-agent extraction algorithms (LangGraph: entity-extraction, relationship-analysis, consolidation, and reflection agents running in parallel) for complex regulatory and technical documents, and ran model validation against an SAP-system-derived ground truth, driving precision/recall from 33%/45% to 93%/90%.
- Engineered a deterministic (code-based) extraction pipeline for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing extraction accuracy toward 100%.
- Designed and implemented a Model Context Protocol (MCP) query service exposing the knowledge graph for millisecond-latency retrieval, integrated directly into Claude Code for agentic development workflows.
- Developed Expert Interface, a graph curation web app (React frontend, Python microservices backend) deployed on Azure, enabling domain experts to validate and approve knowledge-graph changes.
- Delivered as an AI-augmented engineer – leveraging Claude Code, custom skills, and agentic coding workflows for rapid, high-quality development grounded in clean code principles.
- **Tech stack: Python, Docker, Kubernetes, Azure, FastAPI, LangGraph, LangChain, LiteLLM, Neo4j, MongoDB, GraphRAG, RAG, Vector Search, MCP, React, Postgres**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Built and containerized the entire stack with Docker and deployed it to AWS using CDK as Infrastructure as Code, adhering to clean code practices. Version-controlled via GitLab, with CI/CD pipelines for deployment.
- Designed and deployed an agentic AI application using LangGraph to optimize decision-making workflows for vector database queries, general inference, and summarization tasks.
- Developed a React-based front-end and a Python backend with FastAPI, leveraging Postman for API testing.
- **Tech stack: Python, Docker, AWS, CDK, EC2, Lambda, GitLab, GitLab CI/CD, FastAPI, Postman, LangGraph, LangChain, React**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 Leverkusen, Germany

- Automated RAG's Generation and Retrieval evaluation with RAGAS and PytrecEval, comparing model answers to chemist-provided ground truth.
- Built AWS cloud infrastructure with Lambda, SQS, and DynamoDB for PrismaChat RAG evaluation using CDK.
- Benchmarked multilingual embedding models on over 1M data chunks of Covestro's internal chemical data.
- Designed RAG pipelines, focusing on chunking strategies, vector databases, indexing, and fine-tuning models for optimized retrieval.

- **Tech stack: Python, Docker, AWS (S3, EC2, VPC, Lambda, SQS, DynamoDB, CDK), RAGAS, PytrecEval, LangChain, Hugging Face Transformers, PyTorch, TensorFlow, Scikit-Learn, Azure AI Search, Streamlit, Vector Databases**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost, and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity, reaction progression) from time-series process data, evaluating tree-based vs. deep learning approaches for this tabular/temporal hybrid problem.
- Engineered lag features and rolling statistical windows from raw sensor time-series (temperature, reaction time, process variables) to feed the tree-based models, while feeding raw sequences directly into the LSTM.
- Benchmarked model performance against traditional empirical curve-fitting methods, improving early identification of batches likely to deviate from target specification.
- Deployed an interactive Streamlit dashboard exposing model predictions to chemists in real time, letting them simulate parameter changes and see predicted outcomes before running physical batches.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, NumPy, SciPy, Matplotlib, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XG-Boost regressor, FB Prophet, LSTM models for time series analysis for EVs.
- Developed LSTM, achieved 96% test accuracy and strong performance on unseen data.
- **Tech stack: Python, Scikit-Learn, TensorFlow**

PROJECTS

Project: Model Release Pipeline | Python | MLflow | FastAPI | Docker | Kubernetes | GitHub Actions

- MLOps pipeline for a scikit-learn model: MLflow tracking, model packaging, versioning and registry with a model validation gate before promotion, FastAPI service in Docker on a local Kubernetes cluster (kind), GitHub Actions CI/CD with automated testing (pytest), image scan and secure delivery checks, monitoring and observability (latency logging, data-drift report).

TECHNICAL SKILLSET

- **Programming & Version Control:** Python, Git, GitLab
- **Containers, CI/CD & Cloud:** Docker, Kubernetes, GitLab CI/CD, AWS CDK (IaC), Microsoft Azure, AWS (S3, EC2, Lambda, SQS, DynamoDB, VPC)
- **ML & Data Workloads:** Scikit-Learn, XGBoost, PyTorch, TensorFlow, RAGAS, LangChain, LangGraph, Hugging Face Transformers, FastAPI, Postman
- **Databases:** SQL, Postgres, MongoDB, Neo4j, Vector Databases (Qdrant, Pinecone, Azure AI Search)
- **Methodologies:** Agile, software architecture, DevOps practices, AI-assisted development with Claude Code

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering — August 2017 – July 2021
Minor in Computer Science & Engineering
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles

LANGUAGES

English (Business Fluent) | German (B1) | Hindi (Fluent) | Telugu (Native)